

EE1060 Differential Equations & Transform Techniques

I Logistics

- ① Exam Slets:
 03 Feb
 24 Feb } Tue 6:30pm - 8:30pm.
 24 Mar }
 09 Apr } Thu
 30 Apr }

② Attendance :

Taken at random.

If the number of missed roll calls = m
 Penalty = $\begin{cases} 0, & \text{if } m \leq 2 \\ 2(m-2), & \text{if } m > 2. \end{cases}$

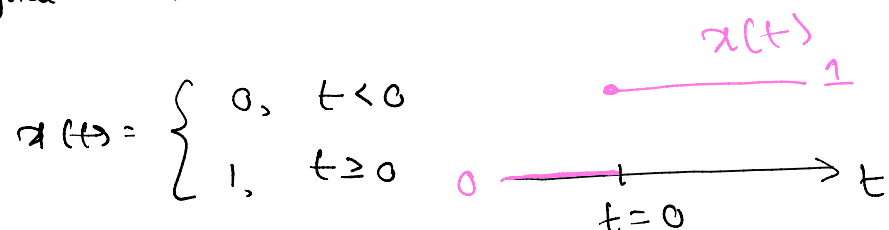
- ③ Google classroom code: um6w2faq

II Motivation

Functions, signals or waveforms appear in a wide variety of engineering applications and carry information about the behaviour of some phenomenon of interest.

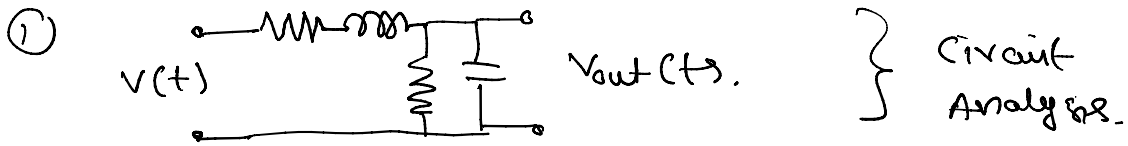
Typically, a signal is described as a function of a time parameter (or space parameter) t .

Signal x ; value $x(t)$ at time t .

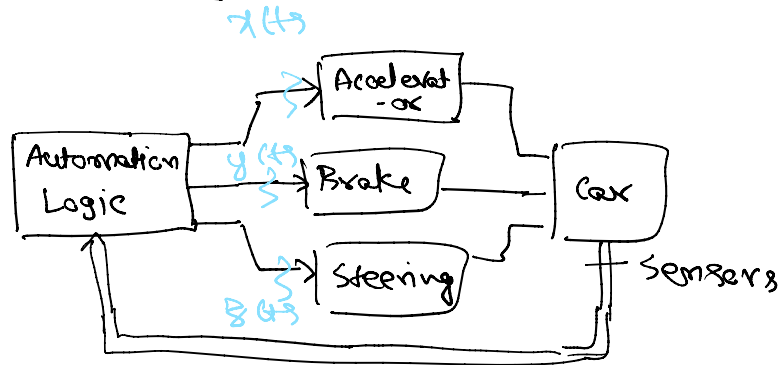


$t=0$

Examples.

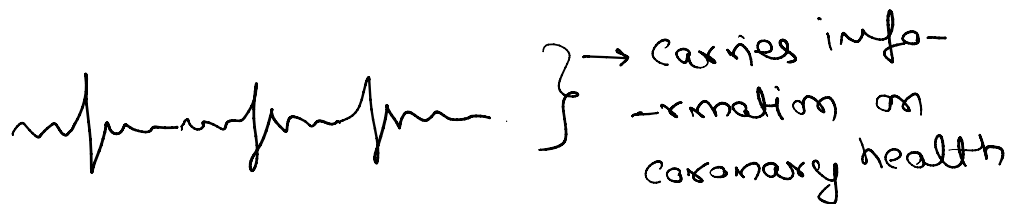


② Control System

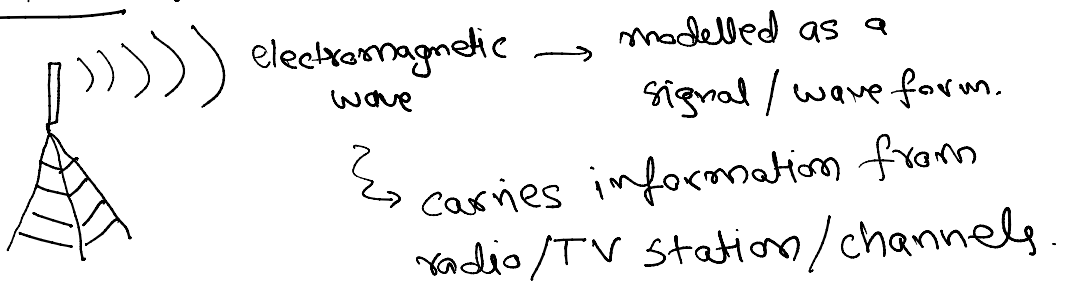


③ Signal Processing

Reading of an electrocardiogram $\rightarrow$ voltage signals that capture the electrical activity of heart.

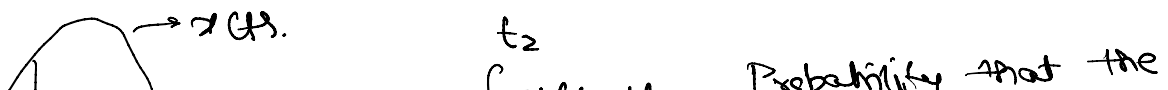


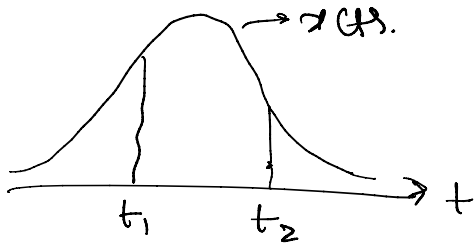
④ Communications



⑤ Probability Theory.

$x(t)$ = probability density function
(Describes a random variable)

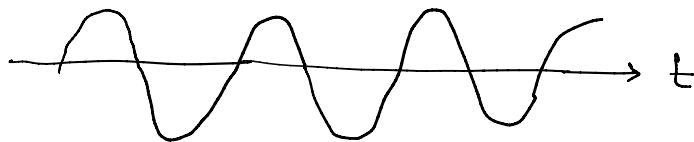




$\int_{t_1}^{t_2} x(t) dt$ = Probability that the random variable takes values in the interval from t_1 to t_2 .

Transform-domain Description of a Signal :

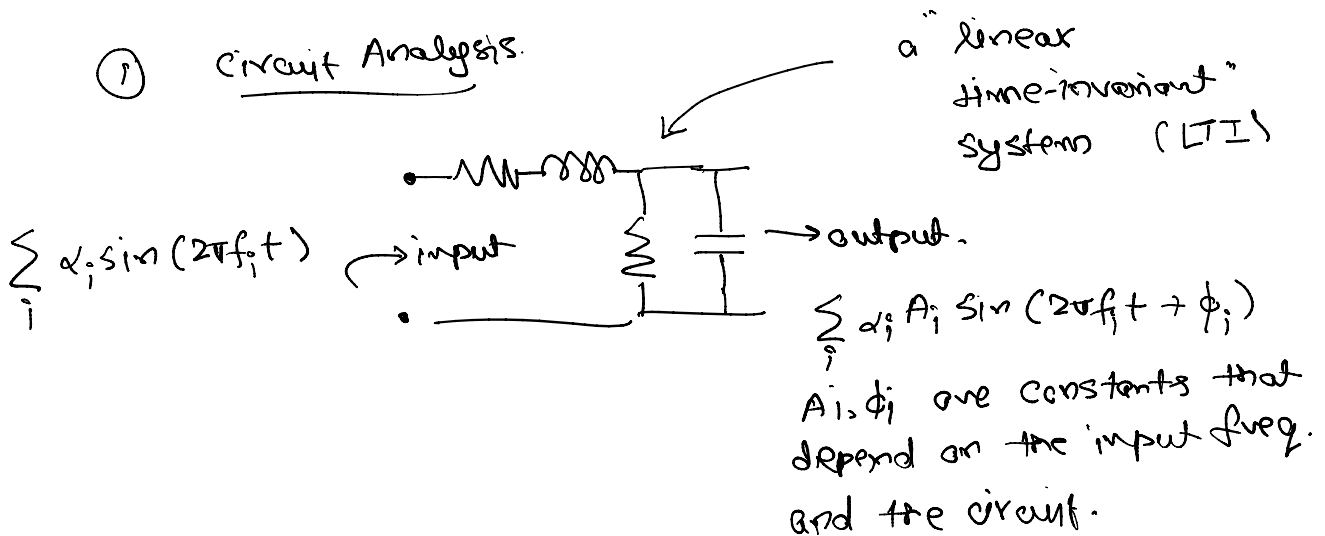
Main idea: We express, either exactly or approximately, the signal of interest to us $x(t)$ as a linear combination of sinusoidal waveforms.



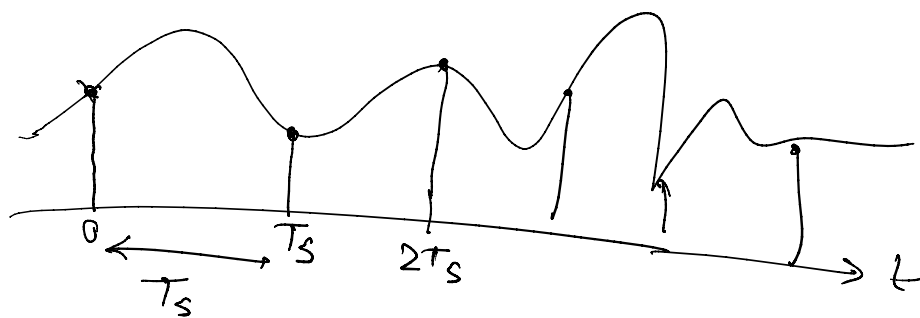
Also known as: frequency-domain analysis, or spectral analysis.

Examples:

① Circuit Analysis.



② Signal Processing.



↳ T_s = sampling interval.

Can we retrieve the value of $x(t)$ at all t , given the values of the samples $x(0)$, $x(T_s)$, $x(2T_s)$, ...
 ↳ Answered by Sampling Theorem.

③ Probability Theory.

$$\frac{X_1 + \dots + X_n}{\sqrt{n}} \longrightarrow \text{Gaussian distribution as } n \rightarrow \infty$$

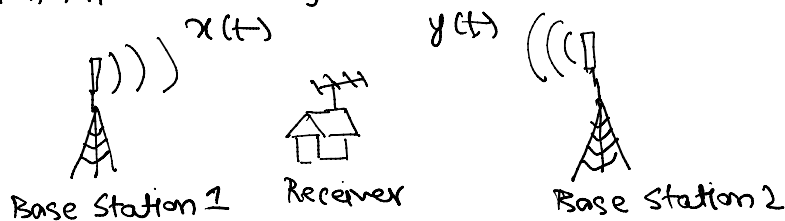
Central limit theorem.

↳ The reasoning / proof uses transform-domain analysis

↳ moment generating function (MGF)

④ Communications.

In practice, multiple wireless transmitters broadcast information simultaneously.

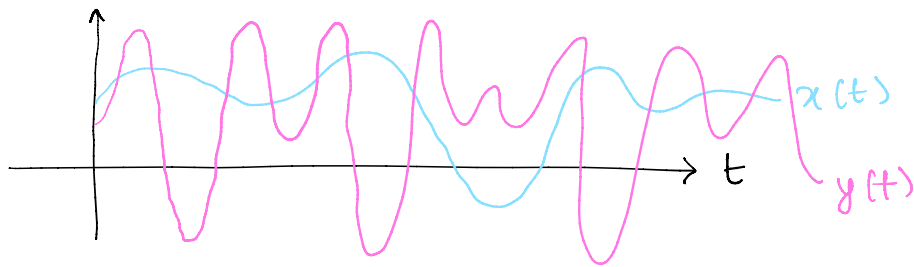


The receiver observes the sum of $x(t)$ and $y(t)$

$$r(t) = x(t) + y(t)$$

the receiver receives

$$x(t) = x(t) + y(t)$$



Can we design $x(t)$ and $y(t)$ such that:

1. they carry the information being broadcast from the two base stations, respectively; and
2. $x(t)$ and $y(t)$ can be separated from each other at the receiver?

Answer: Yes.

A popular solution to this problem is called Frequency-Division Multiple Access (FDMA), where $x(t)$ and $y(t)$ are separated in the frequency domain.

